Subject Description Form

Subject Code	SD5531
Subject Title	Lab Studies in Systems Design
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Lab studies provides an opportunity for students to develop their own future-shaping creative design vision through a series of experiments and research activities. Students are provided with opportunities to be involved in 'real-world' research projects, access to lab equipment which is normally more research-oriented and technologically advanced, and a more 'hands-on' approach. Students will be able to learn through zero-distance contact with more established and experienced researchers in the field. This subject provides students with practical and experimental platforms to experience a hands-on development process for functional systems that interact with people and environments. The main objective is to develop the student's ability to see systems as a network of interacting components or modules and to appreciate the technical characteristics of the different types of components or modules (specifically relating to mechanical, electronic, information and computational aspects), and their implications to overall system performance and impact on people and environments. Potential systems to be explored range from wearables, home automation, remote monitoring, medical devices, personal fabrication, social manufacturing, sustainability, or new modalities in interactions.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional skills</u> a. Demonstrate an ability to form ideas through an experimental process b. Demonstrate an ability to think systemically and to model systems c. Produce and document experiments and research d. Defend work in a critique context e. Carry out a research project with sound time/project management <u>Transferable skills</u> f. Work effectively in groups and collaborate across disciplines g. Appreciate design approaches without a clear solution or problem h. Experience and understand an experimental design approach i. Find creative opportunities in constraints
Subject Synopsis/ Indicative Syllabus	Students will be introduced to: • Data representation and manipulation • Communication protocols • System modelling and sysML • Game and virtual reality applications • Sample applications using intelligent systems

Teaching/Learning Methodology

A successful project will have a substantial research element gathered from case studies, new concepts in the field, user research, and studies on available and emerging technologies.

Activity	Purpose
Lecture	To introduce students to the concepts, theories, and principles of lab studies and the relevant theme/topic.
Seminar	Students, individually or with a group, research an assigned topic, provide a presentation to the class and facilitate the discussion.
Tutorial	To guide students on the development of their experiments' progress, discuss concepts and knowledge of general relevance to support students' experiments individually and in small groups.
Critique	To develop an ability to critique others' work, communicate in a constructive and meaningful manner, and be able to receive feedback from others.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed								
		a	b	c	d	e	f	g	h	i
Initial Experiment Proposal	10%	✓	✓			✓	✓	✓	✓	✓
Interim Presentation	20%	✓	✓		✓	✓	✓	✓	✓	✓
Final Presentation	30%		✓		✓		✓	✓	✓	✓
Experiment Logbook	10%			✓	✓	✓			✓	
Concluding Report	30%		✓	✓	✓		✓	✓	✓	✓
Total	**100%**									

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Initial Experiment Proposal	To evaluate the level of tech sensitivity, depth, and uniqueness of the experiment; the student's ability to develop a lab study project requiring a well-defined but succinct hypothesis and research question that captures the essence of the exploration and leads to an innovation space. A good hypothesis will indicate the value of a design-driven innovation approach.

	Interim Presentation	To evaluate the overall experiment's progress, and student's ability to identify any issues, needs, or creative design opportunities; the experiment process development and project management; evidence of divergent and convergent thinking; a range of concept explorations and quality; visualisation skills; verbal and visual communication of ideas.
	Final Presentation	To evaluate the student's ability to evolve and refine hypothesis and research question as the lab study project progresses with experiments and outcomes; progress through the project to gradually form an idea of what the project's final outputs might be.
	Experiment Logbook	To evaluate the student's learnt ability to record daily progress of experimental and other pertinent information related to the experiment. It should contain sufficient information and commentary to enable the reproduction of experiment results.
	Concluding Report	To evaluate the overall design of the experiment, key stages, expected outcomes, and the ability to answer the initial hypothesis. Lab study projects are not expected to deliver a market-ready design solution, but they are expected to signpost future development opportunities for industrial applications or commercialisation.

Student Study Effort Expected	Class contact	
	▪ Lectures, workshops, and seminars	12 Hrs.
	▪ Tutorials: group and individual	21 Hrs.
	▪ Critiques	6 Hrs.
	Other student study effort:	
	▪ Self-study	26 Hrs.
	▪ Experiment work	52 Hrs.
	Total student study effort	**117 Hrs.**

Reading List and References

- Friedenthal, Sanford, Alan Moore, and Rick Steiner. *A practical guide to SysML: the systems modeling language.* Morgan Kaufmann, 2014.
- Dawson, Michael. *Python programming for the absolute beginner.* Boston, MA: Course Technology, 2010.
- Lutz, Mark. *Learning python: Powerful object-oriented programming.* O'Reilly Media, Inc., 2013.
- Chun, Wesley. *Core python programming.* Prentice Hall Professional, 2001.
- Schmuller, Joseph. *Sams teach yourself UML in 24 hours*. Sams publishing, 2004.
- Scott, Kendall. *Fast Track UML 2.0.* Apress, 2004.
- Delligatti, Lenny. *SysML distilled: A brief guide to the systems modeling language.* Addison-Wesley, 2013.
- White, Elecia. *Making Embedded Systems: Design Patterns for Great Software.* O'Reilly Media, Inc., 2011.

	• Vince, John, and Rae Earnshaw, eds. *Digital media: the future*. Springer Science & Business Media, 2000. • Hansmann, Uwe, Lothar Merk, Martin S. Nicklous, and Thomas Stober. *Pervasive computing: The mobile world*. Springer Science & Business Media, 2003. • Dey, Anind K., Albrecht Schmidt, and Joseph F. McCarthy, eds. *UbiComp 2003: Ubiquitous Computing: 5th International Conference, Seattle, WA, USA, October 12-15, 2003, Proceedings*. Vol. 2864. Springer, 2003. • Liu, Jane W. S. *Real-Time Systems.* Upper Saddle River, NJ: Prentice-Hall, 2000. • Li, Qing, and Caroline Yao. *Real-Time Concepts for Embedded Systems*. CRC Press, 2003.